

30 Days SQL Micro Course Certificate Assignment

1. Total Number of Patients

- Write an SQL query to find the total number of patients across all hospitals.

```
select sum(Patients_Count) from hospital_data;
```

2. Average Number of Doctors per Hospital

- Retrieve the average count of doctors available in each hospital.

```
select Hospital_Name, avg(doctors_count) as avg_doctor from hospital_data  
group by Hospital_Name;
```

3. Top 3 Departments with the Highest Number of Patients

- Find the top 3 hospital departments that have the highest number of patients.

```
select department, sum(patients_count) as total_patients from hospital_data  
group by department  
order by total_patients desc limit 3;
```

4. Hospital with the Maximum Medical Expenses

- Identify the hospital that recorded the highest medical expenses.

```
select Hospital_Name, sum(Medical_Expenses) as highest_bill from hospital_data  
group by Hospital_Name  
order by highest_bill desc limit 1;
```

5. Daily Average Medical Expenses

- Calculate the average medical expenses per day for each hospital.

```
select Hospital_Name, sum(Medical_Expenses)*1.0 / sum(Discharge_Date -  
Admission_Date) as diff_day  
from hospital_data
```

group by Hospital_Name;

6. Longest Hospital Stay

- Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

```
select Hospital_Name, Department, (Discharge_Date - Admission_Date) as long_diff_day  
from hospital_data  
order by long_diff_day desc limit 1;
```

7. Total Patients Treated Per City

- Count the total number of patients treated in each city.

```
select locations, sum(patients_count) AS total_patients  
from hospital_data  
group by locations;
```

8. Average Length of Stay Per Department

- Calculate the average number of days patients spend in each department.

```
select department, avg(Discharge_Date - Admission_Date) avg_stay_days  
from hospital_data  
group by department;
```

9. Identify the Department with the Lowest Number of Patients

- Find the department with the least number of patients.

```
select department, sum(patients_count) as total_patients from hospital_data  
group by department  
order by total_patients  
limit 1;
```

10. Monthly Medical Expenses Report

- Group the data by month and calculate the total medical expenses for each month.

```
select extract(Month from Admission_Date) as Months_no, TO_CHAR(Admission_Date,
'Month') as Months_name, sum(Medical_Expenses) as total_expense
from hospital_data
group by extract(Month from Admission_Date), TO_CHAR(Admission_Date, 'Month')
order by Months_no;
```